INCH-POUND

MIL-D-64022 (AR)
11 September 1991

MILITARY SPECIFICATION

DISTRIBUTION BOX ASSEMBLY: 8438325

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Department and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for a distribution box assembly, hereafter referred to as the D-box. The D-box, is a part of the Vulcan Weapon System M163A1 and M167A1. (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel, General Specification covering the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-L-3661/5	Lampholder, Housing, Indicator Light, Style LH73
MIL-L-3661/12	Lampholder, Lens, Indicator Light, Style LC12

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6110

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STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000	Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8438308	Wiring, Installation of, Distribution Box
8438325	Distribution Box Assembly
8438346	Component Assembly, Slew
8438351	Schematic Diagram, Electrical Dist Box

PACKAGING DATA SHEET

PS8438325 Packaging of Distribution Box Assembly

(Copies of drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 8438325 all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the D-box shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3. Environmental service conditions.

3.3.1 Vibration. The D-box shall be mounted on the vibration table in normal service position. The D-box shall meet the requirements of 3.4 after being subjected to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 55 Hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. The D-box shall not suffer damage and shall meet the requirements of 3.4 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions, along each of three mutually perpendicular axes. The shock pulse shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 $\pm$ 1 milliseconds (ms). The G value of each impulse shall be within $\pm$ 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5ms.

3.3.3 Altitude.

3.3.3.1 Operating. The D-box shall meet the requirements of 3.4 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. The D-box shall meet the requirements of 3.4 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The D-box shall meet the requirements of 3.4 when stabilized thermally at any temperature from -40 deg.F to +155 deg.F.

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3.3.4.2 Nonoperating (storage). The D-box shall not suffer permanent damage when subjected to temperature extreme of -65 deg.F for a period of 24 hours and +160 deg.F for a period of 48 hours (after temperature stabilization), and shall meet the requirements of 3.4 when returned to room ambient.

3.3.5 Humidity (nonoperating). The D-box shall meet the requirements of 3.4 after exposure to an atmosphere with a relative humidity of 95 ±5 percent for a period of not less than 10 days.

3.4 Electrical.

3.4.1 Continuity. Each conductor shall be electrically continuous as specified by drawing 8438308 Wiring, Installation of, Distribution Box.

3.4.2 Functional. The symbols reference in subsequent paragraphs pertain to electrical schematic drawing 8438351.

3.4.2.1 Meters. The D-box shall contain meters to indicate the auxiliary power input voltage and current.

3.4.2.1.1 Volmeter (M2). The volmeter shall indicate zero to 50 Vdc.

3.4.2.1.2 Ammeter (M1). The ammeter shall indicate zero to 100 Adc. The meter shall be a 50mV full scale type using an external shunt resistor.

3.4.2.2 Circuit breakers.

3.4.2.2.1 Sight Gyro power circuit. The gyro power circuit shall contain a 115V, three phase, 400 Hz, 3A circuit breaker (CB4). It shall be possible to reset the circuit breaker from the front panel.

3.4.2.2.2 Sight current generator power circuit. The sight current generator power circuit shall contain a 24 Vdc, 5A circuit breaker (CB1). It shall be possible to reset the circuit breaker from the front panel.

3.4.2.2.3 Inverter power circuit. The inverter power circuit shall contain a 24 Vdc, 15A circuit breaker (CB3). It shall be possible to reset the circuit breakers from the front panel.

3.4.2.2.4 System power circuit. The system power circuit shall contain a 24 Vdc, 15A circuit breaker (CB2). It shall be possible to reset the circuit breakers from the front panel.

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3.4.2.2.5 Servo amplifier power circuits. The servo amplifier power circuits shall contain a 24 Vdc, 35A circuit breaker (CB6, CB7, and CB8) for each of the three servo amplifier power circuits. Overload in any of the three power circuits shall de-energize the turret power relay. The contacts of the circuit breakers shall automatically reset after removal of the overload.

3.4.2.3 NORMAL/STATIC/TEST switch (S1). This switch shall be a three position toggle switch located on the front panel. The switch shall be momentary in the TEST position. The switch shall control the TURRET POWER relay (K4) and the +24 Vdc circuit which feeds the three servo motor brakes in the gun system. Operation shall be as specified in Table I.

TABLE I. Switch positions.

POSITION	TURRET POWER RELAY (K4)	BRAKES
Normal	Coil Circuit CLOSED	Connected
Static	Coil Circuit OPEN	Disconnected
Test	Coil Circuit CLOSED	Disconnected

3.4.2.4 Protective diodes. Diodes shall be provided in the power circuits to prevent the following:

a. Discharge of the system battery into the auxiliary power source. (CR1)

b. Discharge of the gun battery into the auxiliary power source. (CR2)

c. Damage to the 8.84 volt battery tap from application of voltages exceeding 8.84 Vdc. (CR3)

d. Large voltage transients from the gun motor. (CR4)

e. Large voltage transients from the declutching solenoid. (CR5)

f. Large voltage transients from the loading motor. (CR6)

3.4.2.5 Slew control circuits. The D-box shall contain slew control circuits made up of fixed resistors, variable resistors and diodes. The slew control circuits shall control the signals into the azimuth and elevation servo amplifiers in the gun system. The slew circuits shall contain four relays with 24 Vdc coils connected as specified in Table II.

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TABLE II. Slew control circuits.

RELAY	COIL	COIL RETURN
ACQUISITION TIME DELAY (K14)	J1-C	+24 Vdc RETURN
RIGHT LIMIT (K15)	J8-C	+24 Vdc RETURN
LEFT LIMIT (K16)	J8-D	+24 Vdc RETURN
LOWER LIMIT (K13)	J8-J	+24 Vdc RETURN

3.4.2.6 Power control. Two power control relays shall be provided. These relays shall have 24 Vdc coils connected as specified in Table III.

TABLE III. Power control.

RELAY	COIL	COIL RETURN
SYSTEM POWER (K2)	J1-AC	+24 Vdc RETURN
RADAR POWER (K17)	J1-BE	+24 Vdc RETURN

3.4.2.6.1 SYSTEM POWER relay (K2). This relay shall be operated through the control panel connector J1 and shall apply +24 Vdc system battery power to the control panel.

3.4.2.6.2 RADAR POWER relay (K17). This relay shall be operated through the control panel connector J1 and shall supply +24 Vdc system battery power to the radar connector J9.

3.4.2.7 Fire voltage. A fire voltage of +313.5 to +346.5 Vdc shall be provided by a CONVERTER (A3) operating from the 24 Vdc power. The fire voltage shall be delivered to the FIRING VOLTS relay (K6) through a 24.9 ohm, 20 watt resistor. A shunt resistance of one megohm nominal shall be provided across the output of the converter.

3.4.2.8 Hi rate firing. With the connections of Figure 1, the RATE switch in the HI position and the GUN CLEAR switch in the AUTO position, the D-box shall perform as follows:

a. The SYSTEM POWER switch shall energize the SYSTEM POWER relay (K2) and apply power to the GUN POWER switch.

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b. The ACTION switch shall energize the ACTION relay (K3). The GUN POWER switch shall apply power to the fire voltage CONVERTER (A3) and through the contacts of the ACTION relay (K3) to the TRIGGER switch.

c. The TRIGGER switch shall energize the TRIGGER relay (K5). The TRIGGER relay (K5) shall energize the FIRING VOLTS (K6) and HI GUN MOTOR (K8) relays. After 0.87 to 2.1 seconds BURST LIMIT relay (K7) shall energize causing TRIGGER (K5) and HI GUN MOTOR (K8) relays to de-energize. After an additional 0.7 to 1.95 seconds FIRING VOLTS relay (K6) shall de-energize.

d. CLEARING relay (K11) shall energize for 1.0 to 2.23 seconds after TRIGGER relay (K5) de-energizes.

e. Releasing the TRIGGER switch shall de-energize BURST LIMIT relay (K7).

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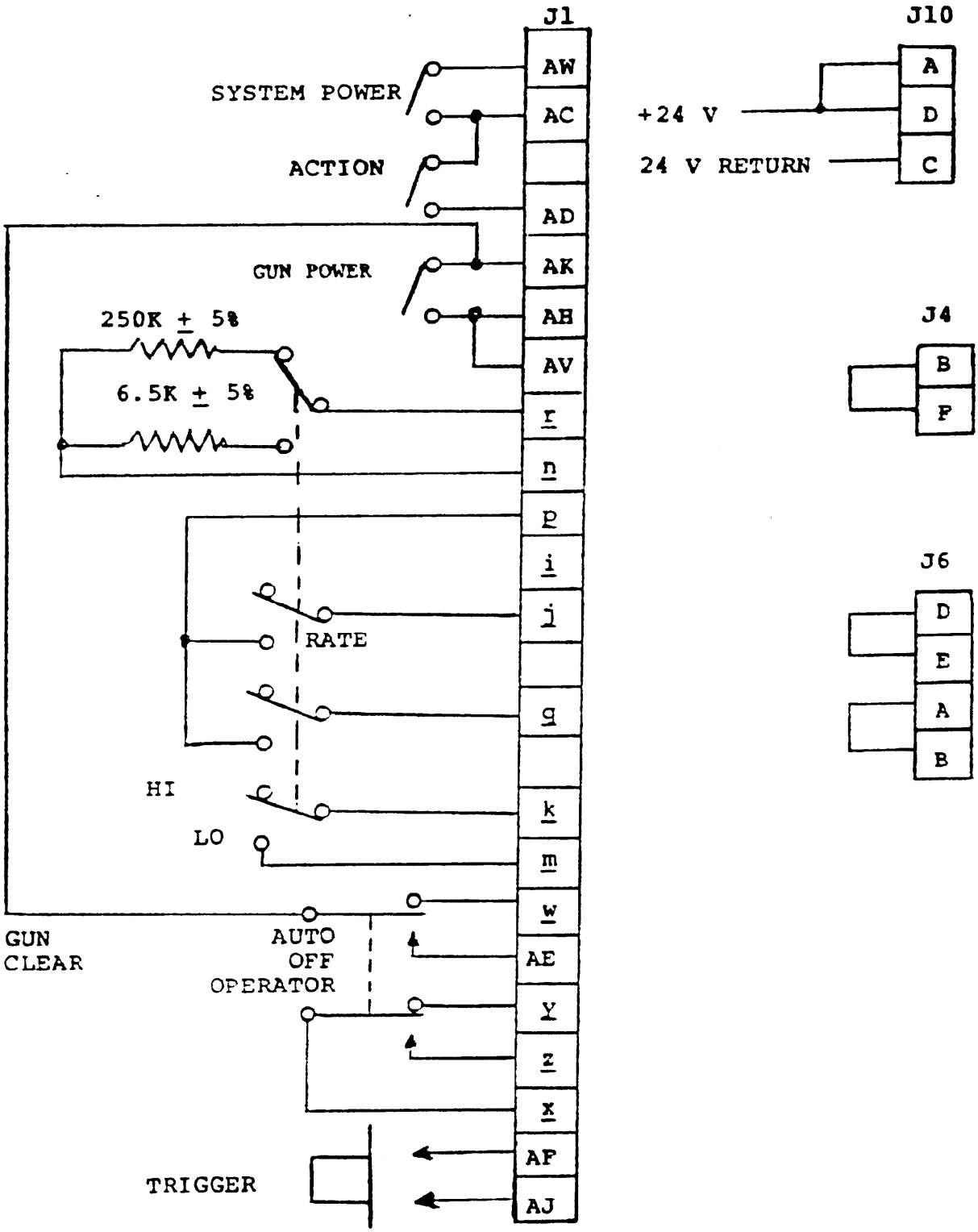


FIGURE 1. Functional connections.

SHEET 1 of 2

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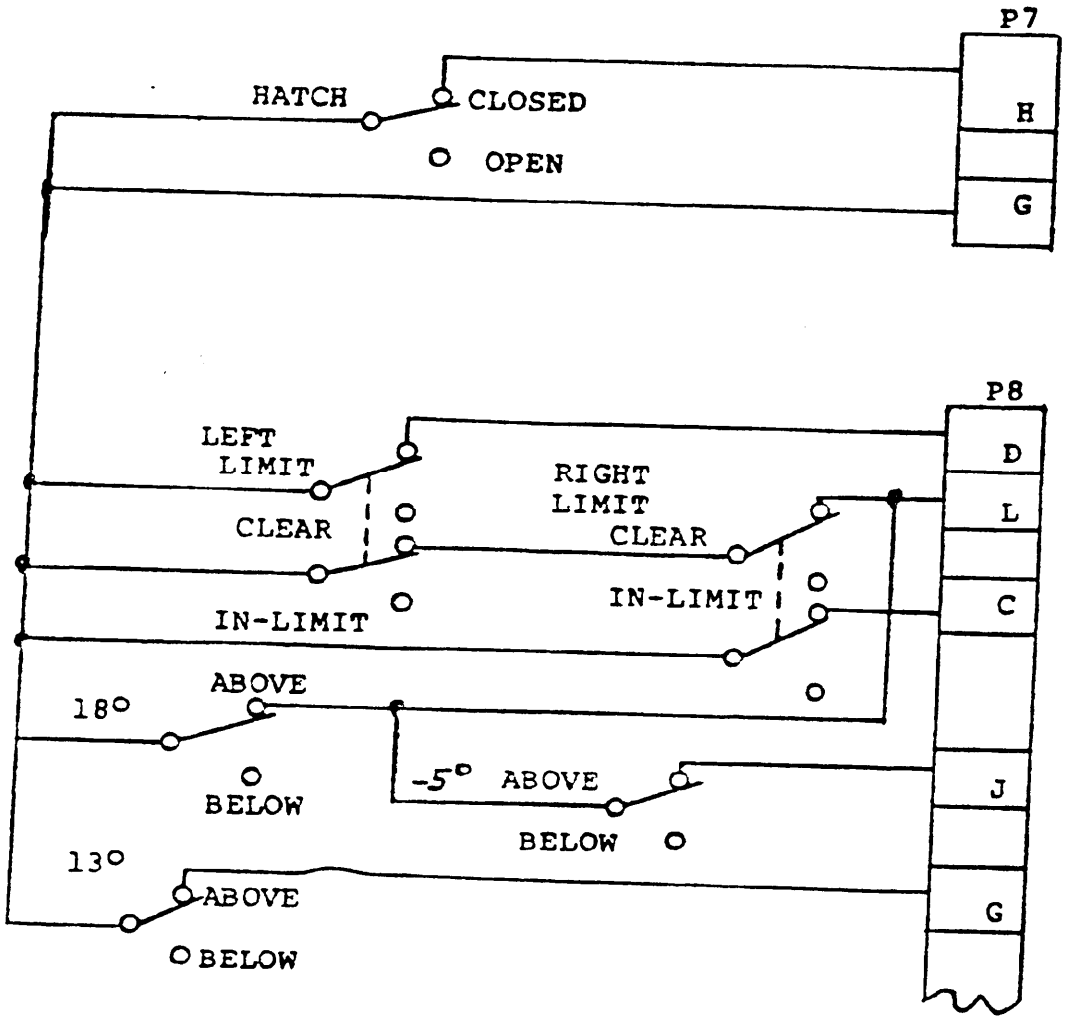


FIGURE 1. Functional connections.

SHEET 2 of 2

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3.4.2.9 Lo rate firing. With the connections of Figure 1, the RATE switch in the LO position and the GUN CLEAR switch in the AUTO position, the D-box shall perform as follows:

a. The SYSTEM POWER switch shall energize the SYSTEM POWER relay (K2) and apply power to the GUN POWER switch.

b. The ACTION switch shall energize the ACTION relay (K3). The GUN POWER switch shall apply power to the fire voltage CONVERTER (A3) and through the contacts of the ACTION relay (K3) to the TRIGGER switch.

c. The TRIGGER switch shall energize the TRIGGER relay (K5). The TRIGGER relay (K5) shall energize the FIRING VOLTS (K6), HI GUN MOTOR (K8), and LO TIME DELAY (K9) relays. The LO TIME DELAY relay (K9) shall energize the LO GUN MOTOR relay (K10). After 26 to 75 milliseconds the BURST LIMIT relay (K7) shall energize causing the HI GUN MOTOR relay (K8) to de-energize.

d. When the TRIGGER switch is released the TRIGGER (K5) and BURST LIMIT (K7) relays shall de-energize immediately. The de-energized TRIGGER relay (K5) causes the CLEARING relay (K11) to energize. The LO TIME DELAY (K9) and LO GUN MOTOR (K10) relays shall de-energize 0.46 to 1.5 seconds after the TRIGGER relay (K5) has de-energized. CLEARING relay (K11) shall de-energize 1.0 to 2.23 seconds after the LO TIME DELAY relay (K9) de-energizes. FIRING VOLTS relay (K6) shall de-energize 0.7 to 1.95 seconds after the LO TIME DELAY relay (K9) de-energizes.

3.4.2.10 Operator clearing. With the connections of Figure 1 and the SYSTEM POWER switch in the ON position and the GUN CLEAR switch in the OPERATOR position, the D-box shall perform as follows:

a. The ACTION switch shall energize the ACTION relay (K3), HI GUN MOTOR relay (K8), and CLEARING relay (K11).

b. Setting the ACTION switch to OFF shall de-energize the ACTION relay (K3) and HI GUN MOTOR relay (K8) immediately. The CLEARING relay (K11) shall de-energize in 0.7 to 1.95 seconds after the ACTION switch is set to OFF.

3.4.2.11 Hatch and elevation protective circuits. The D-box shall contain gun slewing protective circuits. With the connections of Figure 1 and the SYSTEM POWER and ACTION switches ON, these circuits shall operate in conjunction with the hatch switch, the azimuth limit switches and the elevation limit switches to provide the logic Table IV.

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TABLE IV Logic table.

LIMIT CIRCUIT CONDITION						Limit Relays
Hatch	-5 deg.	13 deg.	18 deg.	Right	Left	De-energized
C	C	C	C	C	C	NONE
C	C	C	C	O	O	NONE
C	C	O	O	O	O	NONE
C	O	O	O	O	O	K13
O	O	O	O	O	C	K13 & K15
O	O	O	O	C	O	K13 & K16
O	C	O	O	C	C	NONE
O	C	C	O	C	O	K13
O	C	C	O	O	C	K13
O	C	C	O	C	C	NONE
O	C	O	O	C	O	K13 & K16
O	C	O	O	O	C	K13 & K15
O	C	C	O	O	O	K13, K15 & K16
O	C	C	O	O	O	K13
			C	O	O	NONE

C = Switch Closed O = Switch Open

3.4.2.12 Acquisition time delay relay. The D-box shall contain an acquisition time delay relay K14.

3.4.2.13 Pulse generator. The D-box shall contain a capacitor of 1100 microfarads and 49.9 ohm charging resistor connected through an external rounds counter switch.

3.4.2.14 Filter. The +24 Vdc supplied to pin AU of connector J1 shall contain an LC filter. The LC filter shall consist of a 2.08 millihenery inductor followed by a 100 microfarad capacitor.

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3.5 Design and construction.

3.5.1 General design features.

3.5.1.1 Dimensions. The dimension shall be as follows:

Height	330	mm	(13	in)
Width	254	mm	(10	in)
Length	457	mm	(18	in)

3.5.1.2 Weight. The weight of the D-box shall not exceed 29.9 kg (64 lb).

3.5.1.3 Electrical connectors. The electrical connectors shall be mounted on one 254 x 330 mm end. Pin assignments shall be as specified in Tables V through XIV.

3.5.1.4 Electrical meters. The voltmeter and ammeter shall be mounted on the 254 x 330 mm end opposite that mounting the connectors.

3.5.1.5 Circuit breakers, switch and pressure equalizer valve. These components shall be mounted on one 330 x 457 mm surface.

3.5.1.6 Mounting. Mounting shall be four 3/8 - 24 threaded holes on the same surface as the components listed in 3.5.1.5.

3.6 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

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TABLE V. Control panel connector J1.

PIN	FUNCTION
a	Ground
b	Spare
c	+24 Vdc acquisition time delay pulse from K14
d	Elevation hand control tap
f	Elevation hand control
g	Elevation hand control wiper
h	Ground
i	+24 Vdc firing pulse
j	Burst limit control to K7
k	Burst limit control circuit
m	Low firing rate to K9
n	Firing rate burst limit circuit
p	Burst limit circuit
q	+24 Vdc to trigger relay K5
r	Firing burst limiting input from control panel
s	Elevation hand control
t	Elevation aided laying tach
u	Azimuth aided laying tach
v	Azimuth aided laying tach
w	Gun clear automatic circuit to K9
x	+24 Vdc auto gun clear pulse
y	Gun clear, automatic circuit to K5
z	Gun clear, operator

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TABLE V. Control panel connector J1.(continued)

PIN	FUNCTION
AA	Azimuth hand control tap
AB	Azimuth hand control
AC	+24 Vdc to system power relay K2
AD	+24 Vdc to action relay K3
AE	Gun clear, operator to relay K3
AF	+24 Vdc fire pulse to arming connector J6
AH	+24 Vdc gun power
AJ	+24 Vdc trigger voltage to control panel
AK	Gun clear, 24 Vdc
AL	Servo slew speed control
AM	Azimuth hand control wiper
AN	Azimuth hand control
AP	External range unit
AR	External range unit
AS	Action circuit to relay K3
AT	Action switch command to control panel
AU	+24 Vdc SCG power to control panel
AV	A1J4 interlock jumper
AW	+24 Vdc system power through CB2
AX	+24 Vdc gun power to J4 interlock jumper
AY	Rounds counter
AZ	External range unit

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TABLE V. Control panel connector J1. (continued)

PIN	FUNCTION
BA	+24 Vdc to inverter
BB	+24 Vdc
BC	Ground
BD	Ground
BE	+24 Vdc to radar power relay K17 coil
BF	Ground
BH	Spare
BJ	Elevation 80 degree limit switch
BK	Ground
BL	Ground
BM	115 Vac, 400 Hz, Phase B
BT	115 Vac, 400 Hz, Phase A
BU	115 Vac, 400 Hz, Phase C
BN	Ground
BR	Elevation aided laying tach
BS	115 Vac, 400 Hz, single phase return
BV	115 Vac, 400 Hz, single phase
BP	Ground

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TABLE VI. Inverter converter J2

PIN	FUNCTION
A	N/C
B	N/C
C	115V, 400 Hz, single phase
D	115V, 400 Hz, phase A
E	115V, 400 Hz, phase B
F	115V, 400 Hz, phase C
G	115V, 400 Hz, single phase return
H	Ground
J	N/C
K	+24 Vdc to converter
L	N/C
M	Ground

TABLE VII. Gun drive assembly connector J3.

PIN	FUNCTION
A	Gun motor
B	Brake
C	Ground (Brake return)
D	Ground
E	Firing voltage
F	Declutching Solenoid
G	Ground (Gun motor return)
H	+24 Vdc to Clear and Brake

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VIII. Drum assembly connector J4.

PIN	FUNCTION
A	Rounds counter
B	Loading motor switch
C	Drum full switch
D	+24 Vdc gun power
E	Last round relay
F	Loading unit switches
G	+24 Vdc gun power to K3
H	+24 Vdc to loading motor switch
J	Loading motor
K	Ground (Loading motor return)
L	Rounds counter capacitor
M	Ground
N	Rounds counter charging voltage
P	Battery heaters

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TABLE IX. Test connector J5.

PIN	FUNCTION
A	+8.84 Vdc low gun motor battery voltage
B	Gun motor voltage from relays K8 to K10
C	+24 Vdc when K7 energized
D	+24 Vdc when trigger is on
E	Other side of arming connector
F	O Vdc when K7 energized
G	Coil of last round relay K12 (nonfunctional)
H	+24 Vdc when gun motor brake energized
J	+24 Vdc to energize declutching solenoid
K	+24 Vdc when low limit relay K13 energized
L	+24 Vdc when right limit relay K15 energized
M	+24 Vdc when left limit relay K16 energized
N	+8.84 Vdc low gun motor voltage cathode A1CR3
P	+24 Vdc radar power
R	+24 Vdc battery charging voltage on CR2 Cathode
S	Fire voltage +330 Vdc during fire
T	+24 Vdc when turret power relay K4 energized
U	115 Vac, 400 Hz, phase B
V	115 Vac, 400 Hz, phase A
W	115 Vac, 400 Hz, phase C
X	+24 Vdc when loading relay K1 energized
Y	+24 Vdc system battery voltage
Z	+24 Vdc gun motor battery voltage

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TABLE IX. Test connector J5.(Continued)

PIN	FUNCTION
a	+24 Vdc when ATD relay K14 energized
b	Relay K3 +24 Vdc to control panel
c	115 Vac, 400 Hz, single phase
d	115 Vac, 400 Hz, single phase return
e	Ground
f	Last round relay K12 coil (nonfunctional)
g	Transistor Q1 base circuit
h	+24 Vdc when relay K8 energized
i	+24 Vdc to loading motor
j	+24 Vdc when gun motor relay K10 energized
k	+24 Vdc when system power relay K2 energized
m	+330 Vdc, converter output
n	O Vdc when clearing relay K11 energized
p	O Vdc when firing volts relay K6 energized
q	O Vdc when low rate time delay relay K9 energized
r	+24 Vdc when last round relay K12 energized
s	+24 Vdc when trigger relay K5 energized
t	+24 Vdc power to time delay circuits from K5

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TABLE X. Arming connector J6.

PIN	FUNCTION
A	Remote arming
B	Remote arming
C	Ground
D	+24 Vdc fire pulse from J1
E	+24 Vdc fire pulse

TABLE XI. Connector J7.

PIN	FUNCTION
A	External range unit
B	External range unit
C	External range unit
D	Ground
E	+24 Vdc (External power)
F	Ground (External power)
G	Gun position and hatch switches
H	Gun position and hatch switches
I	Spare

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TABLE XII. Connector J8.

PIN	FUNCTION
A	Elevation servo signal
B	Elevation servo signal
C	Azimuth limit switches
D	Azimuth limit switches
E	Signal ground (Elevation servo)
F	Elevation and azimuth limit switches
G	Elevation limit switches
H	Signal ground (Azimuth servos)
J	Elevation - 5 degree limit switch
K	Azimuth tach
L	Elevation and azimuth limit switches
M	Elevation tach
N	Elevation limit switches
O	Spare
P	Azimuth servo No. 1 power
R	Ground
S	Azimuth servo No. 2 power
T	Ground
U	Ground
V	Elevation servo power
W	Test connection to elevation servo amp
X	Azimuth tach
Y	Spare
Z	Elevation tach

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TABLE XII. Connector J8. (Continued)

PIN	FUNCTION
a	Spare
b	Elevation limit switches
c	Elevation 13 degree limit switch
d	Servo motor brake
e	Ground
f	Ground
g	Ground
h	Azimuth servo No. 1 signal
j	Azimuth servo No. 1 signal
k	Elevation 80 degree limit switch

TABLE XIII. Radar connector J9.

PIN	FUNCTION
A	Radar power
B	Ground
C	Ground

TABLE XIV. Battery connector J10.

PIN	FUNCTION
A	+24 Vdc gun motor voltage
B	+8.84 Vdc low gun motor voltage
C	Ground
D	+24 Vdc system power

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50deg to +90deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

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4.3 First article inspection.

4.3.1 First article submission . The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) distribution boxes for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table XV (Functional Test) and Table XVI (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogenous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any pri or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

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4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table XV shall be performed on a 100% basis.

TABLE XV. Functional Tests
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE	N/A
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.4.2	4.6.3
MINOR:	NONE	N/A

4.5 Special sampling. One D-box out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table XV and shall then meet the tests in Table XVI. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XVI. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity (nonoperating)	3.3.5	4.7.1.5
Minor:	None	N/A

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements of MIL-I-45607.

4.6.2 Contractor furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment esigns shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2 the test equipment listed in Table XVIII (Test Equipment list) shall be used in the performance of the applicable tests of 4.6.

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TABLE XVIII. Test equipment list.

ITEM		DESCRIPTION
1.	Power supply	Nominal value of 24 Vdc and 40 vdc; 0-40 Vdc, adjustable voltage. 0-20 Adc, adjustable current limit. Load regulation 1% or less. Less than 5% ripple
2.	Power supply	Nominal value of 5 Vdc; 0-10 Vdc adjustable voltage. Load regulation 1% or less from 10 to 50 milliamperes. Less than 5% ripple.
3.	Power supply	Nominal value of 24 Vdc; 0-30 Vdc adjustable voltage. 0-1.5 Adc. Load regulation 1% or less. Less than 5% ripple.
4.	Volt-ohmeter	DC voltage accuracy and resolution of 1% or better from 1 to 350 Vdc. Input impedance not less than 10 megohms when measuring 300 to 350V. Resistance accuracy and resolution of 0.2% or better from one ohm to 500 K ohm. Hi/Lo voltage ohmeter.
5.	Oscilloscope	Input impedance of one megohm or greater. Single sweep capability.
6.	Oscilloscope probe	Attenuation of 100X. Input resistance of 10 megohms.
7.	Audio oscillator	Capable of producing a frequency of 1000 Hz. Output impedance of 50 ohms. Output voltage into 50 ohms, not less than 3.0 volts rms.
8.	Test fixture	See Figure 2 for schematic and Table XIX for the parts list.

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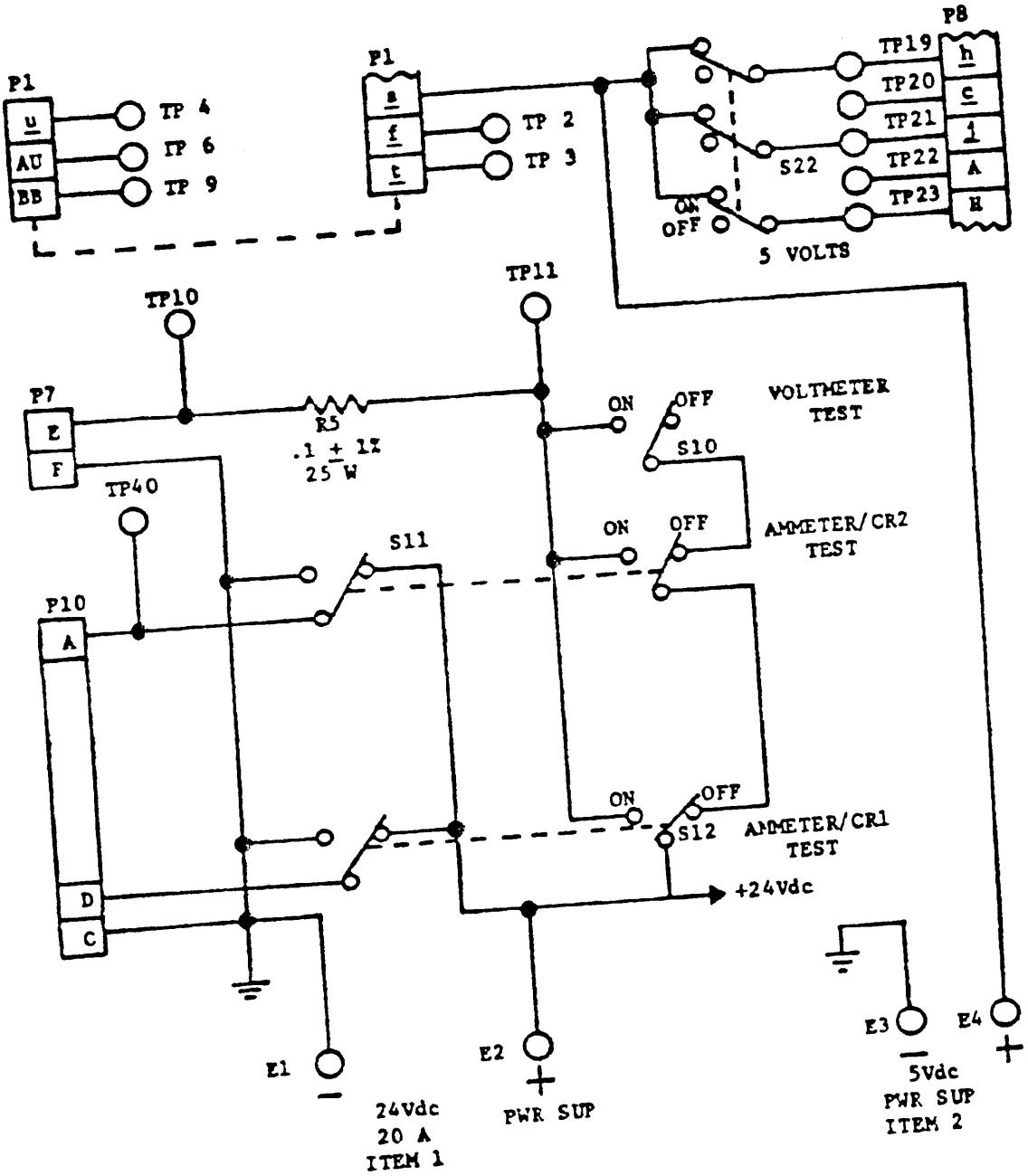


FIGURE 2. Test fixture.

Sheet 1 of 7

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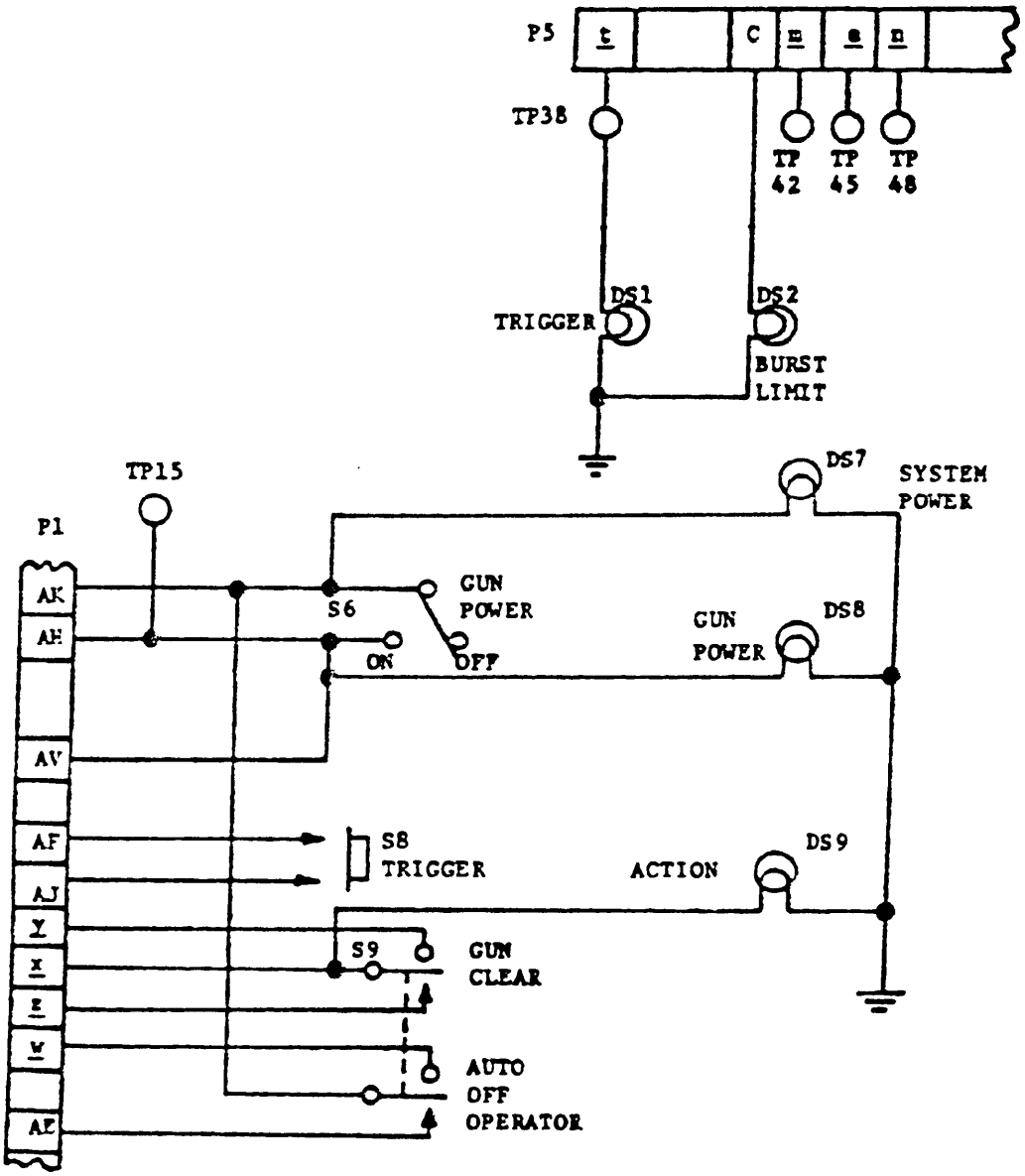


FIGURE 2. Test fixture.

SHEET 2 of 7

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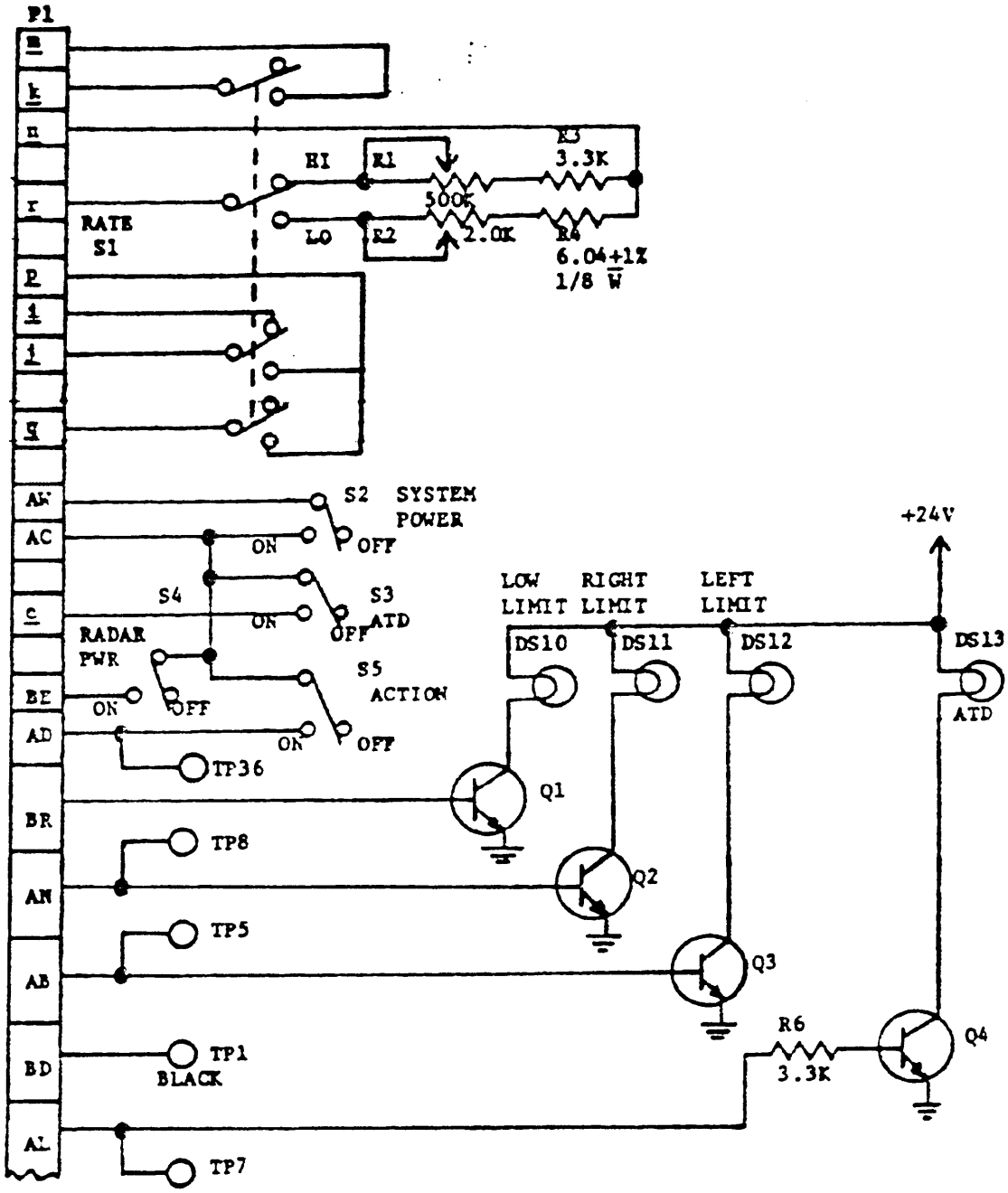


FIGURE 2. Test fixture.

SHEET 3 of 7

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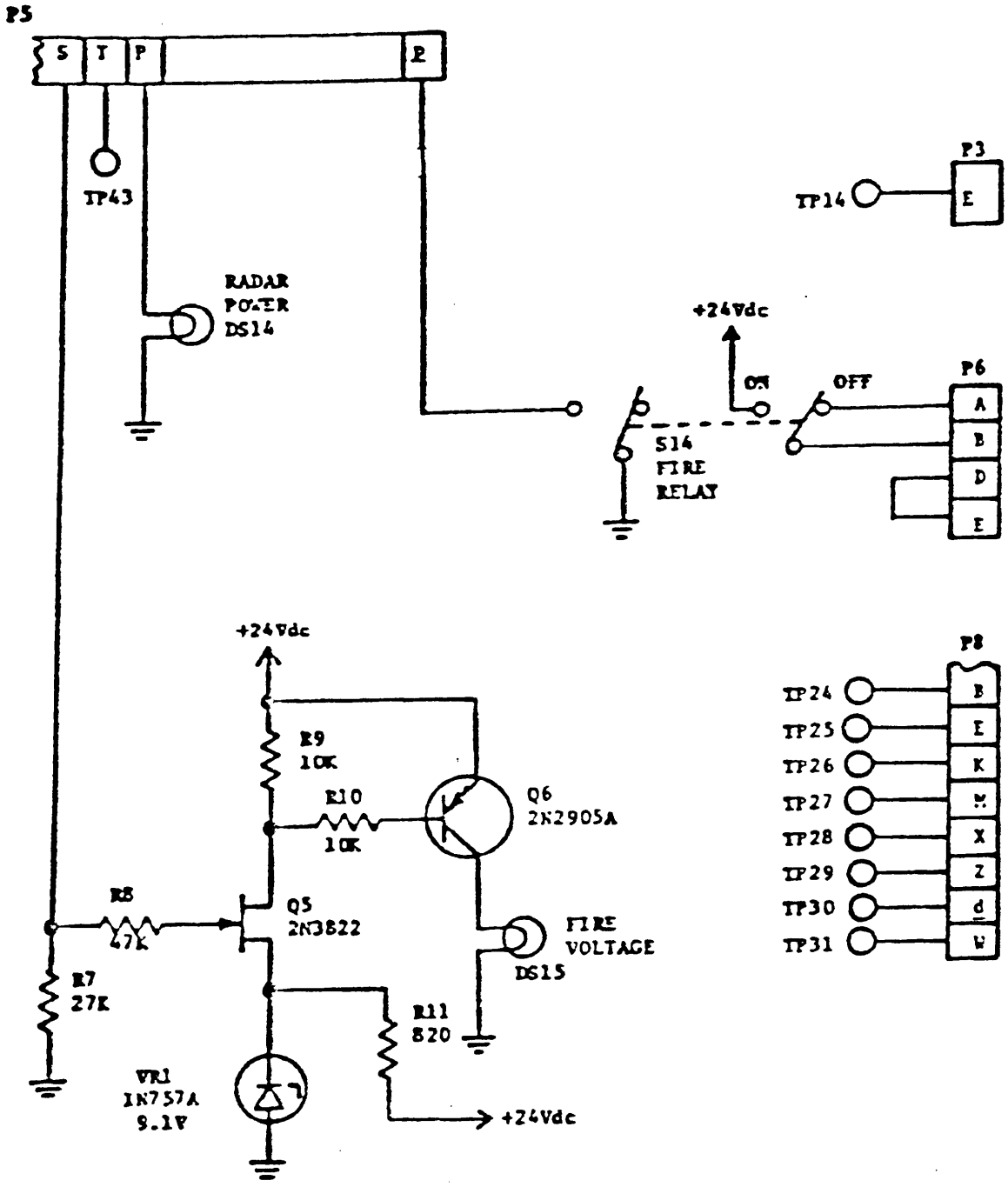


FIGURE 2. Test fixture.

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and the wreckage remain undisturbed until movement is authorized by the

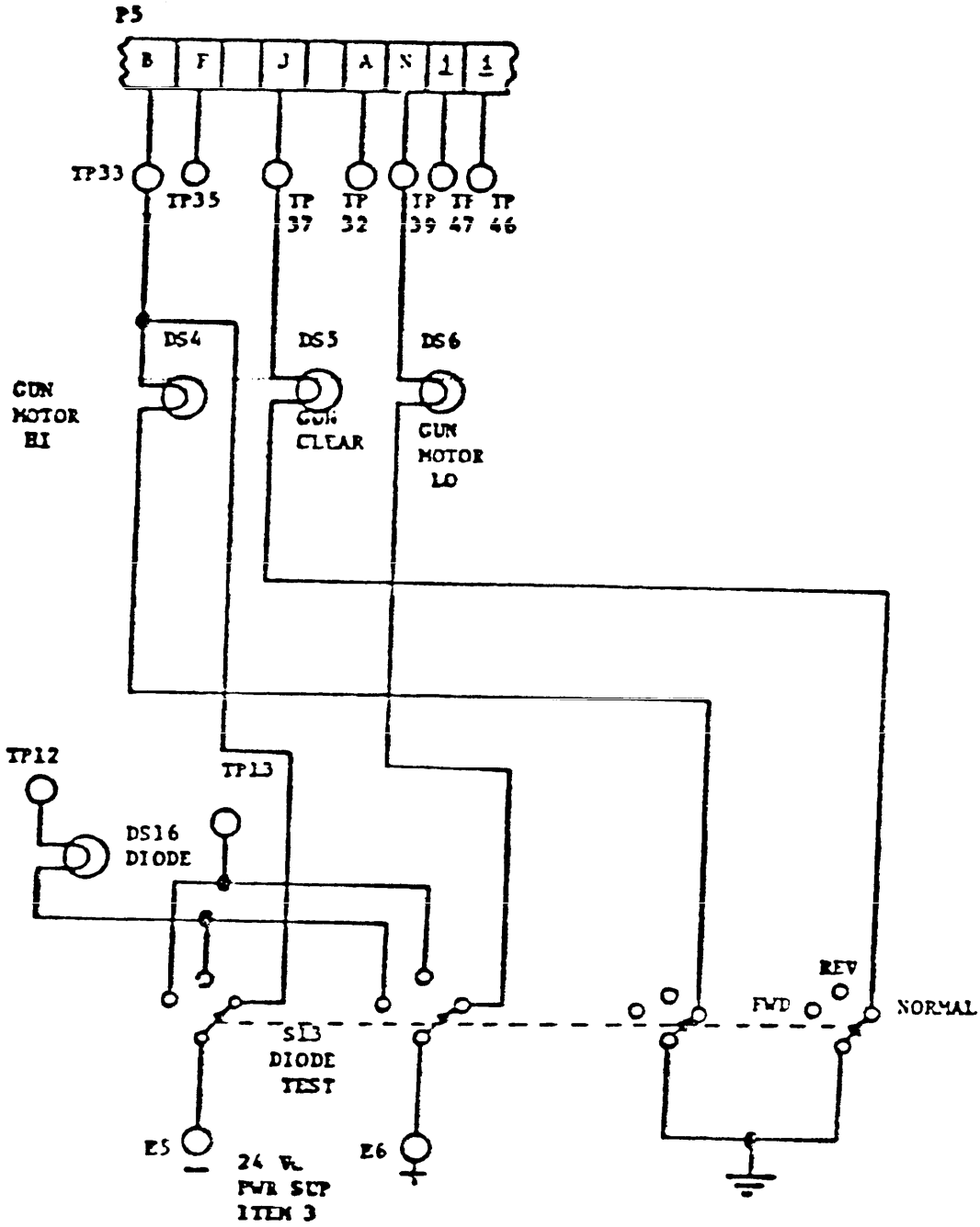


FIGURE 2. Test fixture

SHEET 5 of 7

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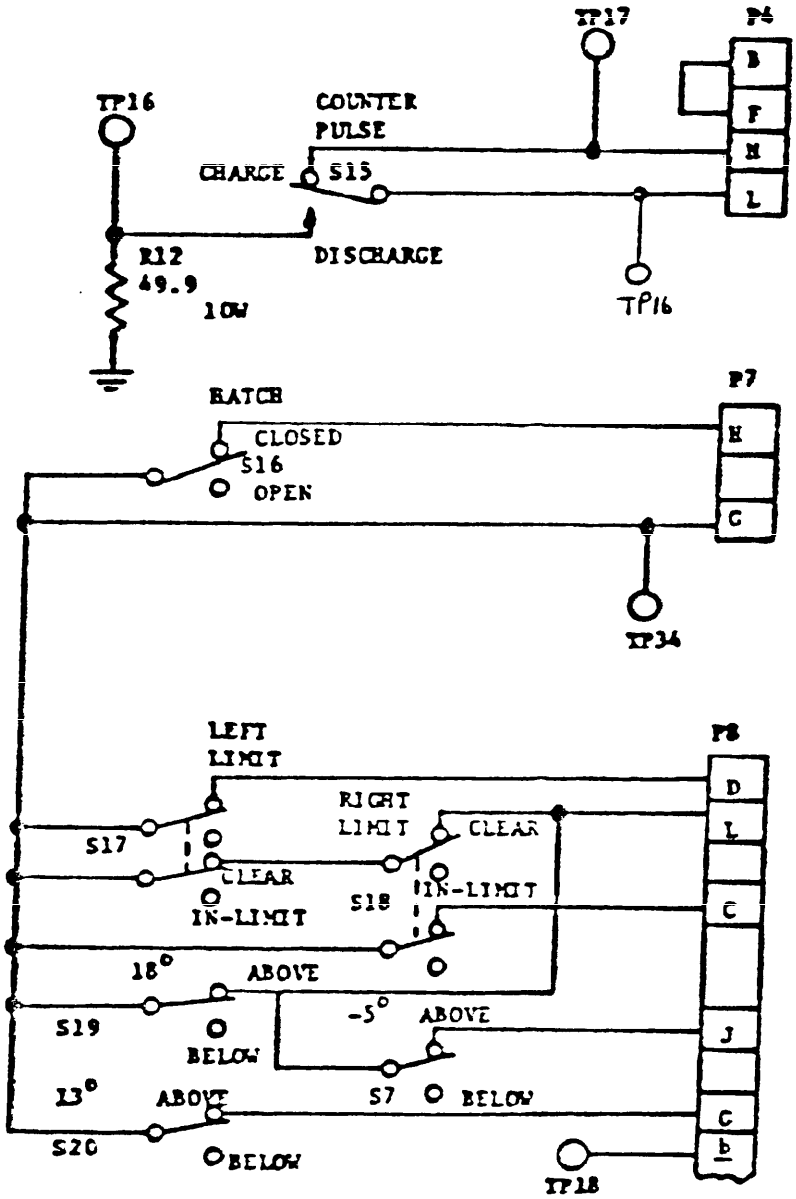


FIGURE 2. Test fixture

SHEET 6 of 7

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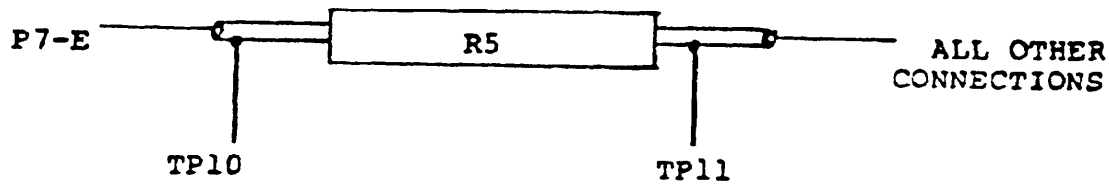
NOTES:

1. ADJUST R1 FOR A TOTAL RESISTANCE OF 250K ±5% ACROSS R1 AND R3.

2. ADJUST R2 FOR A TOTAL RESISTANCE OF 6.5K ±5% ACROSS R2 AND R4.

3. ALL RESISTORS ARE 1/4 WATT AND ±5% UNLESS OTHERWISE NOTED.

4. WIRE R5 AS SHOWN BELOW:



5. ALL WIRING TO P7, P10 AND THEIR ASSOCIATED COMPONENTS IS TO BE #12 AWG.

6. SWITCH S15 IS TO HAVE LOW CONTACT BOUNCE. A ROTARY TYPE IS RECOMMENDED.

FIGURE 2. Test fixture.

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TABLE XIX. Test fixture parts.

DESIGNATION	DESCRIPTION	SUGGESTED PART NUMBER
DS1 thru DS16	Lamp miniature, 28V, .04A	MS25237-327
P1	Connector, Plug	MS3106F40-56P
P3	Connector, Plug	MS3106A32-15P
P4	Connector, Plug	MS3106F28-2P
P5	Connector, Plug	MS3126F20-41P
P6	Connector, Plug	MS3126F14-5P
P7	Connector, Plug	MS3106F24-11S
P8	Connector, Plug	MS3106F32-7P
P10	Connector, Plug	MS3106F36-5S
Q1 thru Q4	Transistor, NPN, silicon	2N2219A
Q5	Transistor, field effort, N-channel	2N3822
Q6	Transistor, PNP, silicon	2N2905A
R1	Resistor, variable, 500K	RJR12FP504M
R2	Resistor, variable, 2.0K	RJR12FP202M
R3 & R6	Resistor, 3.3K ±5%, 1/4 watt	RCRO7G332JM
R4	Resistor, 6.04K ±1% 1/8 watt	RNC55H6041FS
R5	Resistor, .1 ohm ±1% 25 watt	RE70GR100
R7	Resistor, 27K ±5%, 1/4 watt	RCRO7G273JM
R8	Resistor, 47K ±5%, 1/4 watt	RCRO7G473JM
R9 & R10	Resistor, 10K ±5%, 1/4 watt	RCRO7G103JM
R11	Resistor, 820 ±5%, 1/4 watt	RCRO7G821JM
R12	Resistor, 49.9 ±5%, 10 watt	RWR78S49R9FM

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TABLE XIX. Test fixture parts. (continued)

DESIGNATION	DESCRIPTION	SUGGESTED PART NUMBER
S1	Switch, toggle, 4PDT, 28 Vdc, 1A	MS24524-23
S2,S3,S4,S5,S6 S7,S10,S16,S19, S20	Switch, toggle, SPST, 28 Vdc, 5A	MS24523-22
S8	Switch, pushbutton, SPST, N.O. 28 Vdc, 5A	M8805/96-001
S9	Switch, toggle, 2 pole, 3 pos one momentary, 28 Vdc, 5A	MS24524-31
S11 & S12	Switch, toggle, DPDT, 28 Vdc, 20A	MS24524-23
S13	Switch, rotary, 4 pole, 3 pos, non-shorting	M3786/13-0022
S14	Switch, toggle, DPDT, 28 Vdc, 5A	MS24524-23
S15	Switch, rotary, momentary non-shorting	Centralab 1451
S17 & S18	Switch, toggle, DPST, 28 Vdc 5A	MS24524-22
S22	Switch, toggle, 4PST, 28 Vdc 5A	MS24524-24
TP's	Jacks, insulated, banana	HH Smith Type 1508
VR1	Diode voltage regulating, 9.1V ±5%	1N757A
XDS1, XDS2	Lampholder (MIL-L-3661/5) White lens (MIL-L-3661/12)	LH73/1 LC12WT2
XDS4,XDS5,XDS6 XDS13, XDS14	Lampholder (MIL-L-3661/5) Green lens (MIL-L-3661/12)	LH73/1 LC12GT2
XDS7,XDS8,XDS9, XDS15	Lampholder (MIL-L-3661/5) Red Lens (MIL-L-3661/12)	LH73/1 LC12RT2
XDS10,XDS11, XDS12	Lampholder (MIL-L-3661/5) Yellow lens (MIL-L-3661/12)	LH73/1 LC12YT2

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with general requirements and test methods in MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514 Procedure I, Category I of MIL-STD-810 except that:

(1) The frequency range shall be 10 to 55 Hz.

(2) The range shall be swept linearly for two hours per axis at a half amplitude excursion of .016 inch.

(3) The sweep rate shall be from 10 to 55 Hz for approximately 10 minutes.

After vibration, the D-box shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the D-box shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (nonoperating and operating). This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the operating altitude shall be limited to 10,000 feet (20.6 inches hg). Operation of Steps 2 and 4 shall be the performance tests specified in 4.7.3 except that the tests of 4.7.3.4 and 4.7.3.5 are not required.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and nonoperating). This test shall be performed as specified in Procedure I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and nonoperating). This test shall be performed as specified in Procedure I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational check out shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.5 Humidity (nonoperating). This test shall be performed in accordance with Procedure II, Method 507, of MIL-STD-810. The D-box shall be tested in accordance with 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance. The selected D-box shall be tested as follows:

4.7.3.1 Condition of operation. Connect the test fixture, item 8, the D-box under test, and connect power supplies, item 1, 2 and 3 to the test fixture, as shown in Figure 3. Connect the voltmeter, item 4 and the oscilloscope, item 5, to the test fixture as directed during the testing.

4.7.3.2 Preset controls. All controls are located on the test fixture, item 8, and the power supplies, item 1, 2 and 3, with the exception of the NORMAL/STATIC/TEST switch (S1) which is on the D-box under test. Preset the NORMAL/STATIC/TEST switch to NORMAL. Preset all ON-OFF test set switches to OFF. Preset the remaining switches as listed below:

RATE to LO

DIODE TEST to NORMAL

HATCH to CLOSED

LEFT LIMIT to CLEAR

RIGHT LIMIT to IN LIMIT

18 degrees to ABOVE

13 degrees to ABOVE

-5 degrees to ABOVE

GUN CLEAR to OFF

Insure that circuit breakers, CB1 thru CB4, on the D-box are on.

Turn on power supply, item 2,, and set the voltage to 5.0 $\pm$ 0.5 Vdc. Turn on power supply, item 3, and set the voltage to 24 $\pm$1 Vdc.

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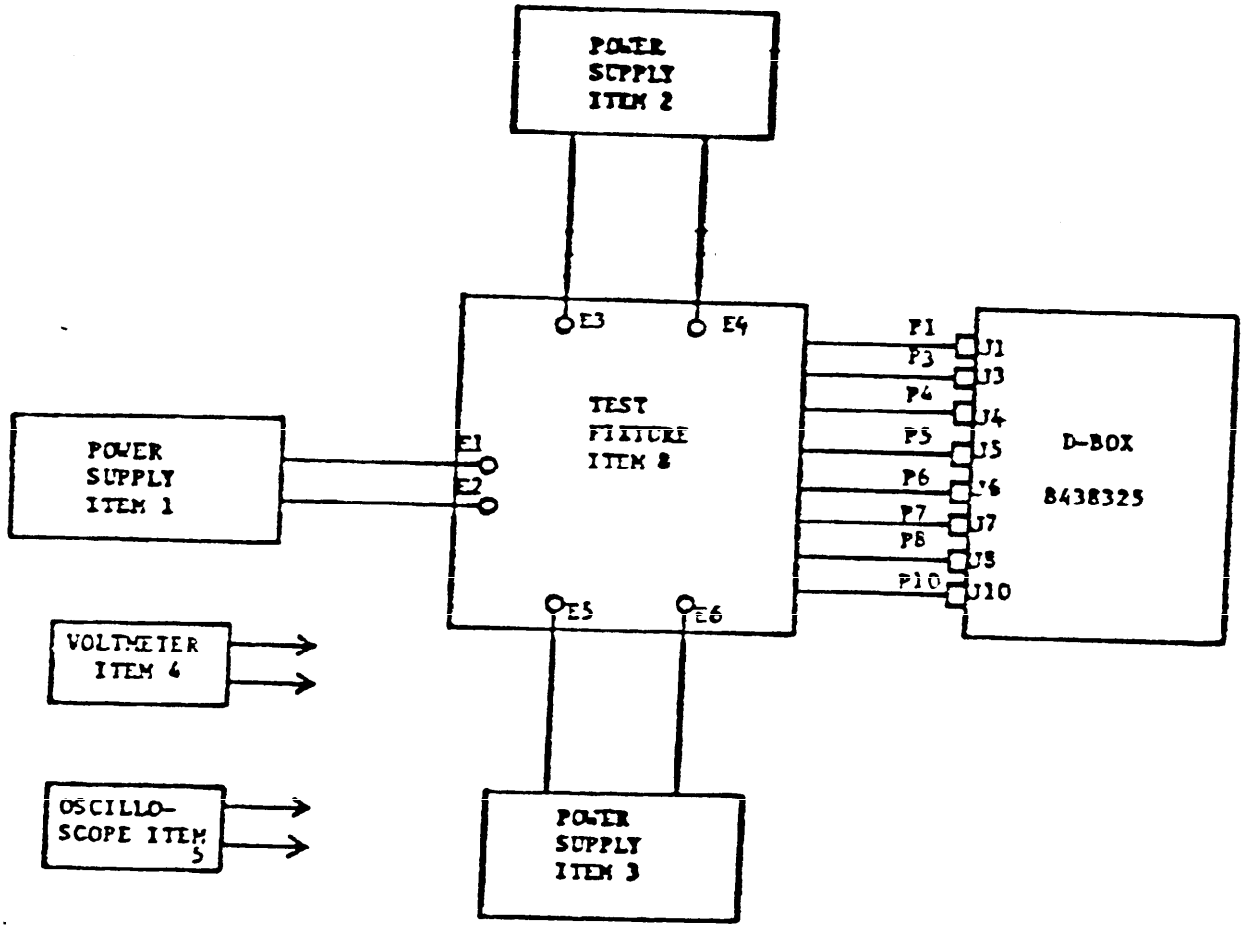


FIGURE 3. D-box test setup.

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WARNING

TEST POINTS TP14 AND TP42 MAY HAVE

HIGH VOLTAGE ON THEM AND COULD CAUSE

SHOCK. USE WITH CAUTION.

4.7.3.3 Meters and diodes CR1 and CR2. Connect the voltmeter, item 4, to TP 40 and the return to TP1. Turn on and adjust the power supply, item 1, until the voltmeter indicates 40.0 ± 0.1 Vdc. The D-box voltmeter (M2) must indicate zero volts. Place the test set VOLTMETER TEST switch to ON. The D-box voltmeter (M2) must indicate 40 ±1 volts. Operate the VOLTMETER TEST switch to OFF. Reduce the power supply, item 1, voltage and current limit to zero. Connect the voltmeter, item 4, to TP11 and the return to TP10. Perform the following steps:

a. Set the AMMETER/CR1 TEST switch to ON.

b. Adjust the power supply, item 1, to 24 ±1 Vdc. Slowly adjust the power supply current limit control for a reading of 2.0 ±0.01 Vdc on voltmeter, item 4.

c. The D-box ammeter (M1) must read 20 ±1 amperes.

d. Set the AMMETER/CR1 TEST switch to OFF and set the AMMETER/CR2 TEST switch to ON.

e. Adjust power supply, item 1, current limit control, as necessary to maintain 2.0 ±0.01 Vdc on the voltmeter, item 4.

f. The D-box ammeter (M1) must read 20 ± 1 amperes.

g. Set the AMMETER/CR2 TEST switch to OFF.

4.7.3.4 Circuit breakers. Visually observe the front panel of the D-box and verify that the circuit breakers are marked in accordance with Table XX.

TABLE XX. Circuit breakers.

CIRCUIT BREAKER	CURRENT RATING (A)
Sight gyro power (CB4)	3
SCG power (CB1)	5
Inverter power (CB3)	15
System power (CB2)	15

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4.7.3.5 NORMAL/STATIC/TEST switch (S1). Verify proper operation of the NORMAL/STATIC/TEST switch using the volt-ohmmeter, item 4. Measure the resistance between test points specified in Table XXI. Continuity shall be indicated by a reading of less than 2.5 ohms.

TABLE XXI. Switch verification.

SWITCH POSITION	TP	TP	CONDITION
NORMAL	43	36	Continuity
STATIC	43	36	100 ohms min
TEST	43	36	Continuity
NORMAL	30	34	Continuity
STATIC	30	34	Open
TEST	30	34	Open
NORMAL	31	34	Open
STATIC	31	34	Open
TEST	31	34	Continuity

Return the NORMAL/STATIC/TEST switch (S1) to NORMAL.

4.7.3.6 Protective diodes With the test point connections and DIODE TEST switch positions of Table XXII, the DIODE indicator lamp condition must be as specified in the table.

TABLE XXII. Diode verification.

DIODE	CONNECT TP	DIODE TEST SW	DIODE LAMP
CR3	32 to 12 and 39 to 13	Fwd	On
CR3	32 to 12 and 39 to 13	Rev	Off
CR4	45 to 12 and 33 to 13	Fwd	On
CR4	45 to 12 and 33 to 13	Rev	Off
CR5	45 to 12 and 37 to 13	Fwd	On

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TABLE XXII. Diode verification.(continued)

DIODE	CONNECT TP	DIODE TEST SW	DIODE LAMP
CR5	45 to 12 and 37 to 13	Rev	Off
CR6	45 to 12 and 46 to 13	Fwd	On
CR6	45 to 12 and 46 to 13	Rev	Off

Remove connections between test points. Return DIODE TEST switch to NORMAL.

4.7.3.7 Slew control circuits. Connect the volt-ohmmeter, item 4, as indicated in Table XXIII. Using the high or low ohmmeter resistance range indicated and the test fixture, item 8, switch positions indicated, observe the designated resistance readings.

TABLE XXIII. Resistance verification.

OHMMETER					
+LEAD TP	-LEAD TP	HI LO SW	RESISTANCE (OHMS)	SET SLEW CCA (8438346) POTS FOR PROPER INDIC	TEST SET SWITCHES
24	3	LO	200K to 240K		
24	27	LO	275K to 315K	R32	
21	4	LO	430K to 510K		
25	18	LO	9.27K to 9.47K	R24	
25	22	LO	9.27K to 9.47K	R23	
20	2	LO	3.2K to 3.4K	R34	
22	29	LO	275K to 315K	R31	
23	21	LO	9.27K to 9.47K	R20	
23	19	LO	9.27K to 9.47K	R19	
19	28	LO	220K to 240K	R29	
21	26	LO	220K to 240K	R30	
23	26	HI	> 100K		

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TABLE XXIII. Resistance verification. (continued)

OHMMETER					
+LEAD TP	-LEAD TP	HI LO SW	RESISTANCE (OHMS)	SET SLEW CCA (8438346) POTS FOR PROPER INDIC	TEST SET SWITCHES
23	28	HI	> 100K		
26	23	HI	< 10K		
28	23	HI	< 10K		
25	27	HI	> 100K		
25	29	HI	> 100K		
27	25	HI	< 10K		
29	25	HI	< 10K		
23	7	HI	11.5K to 13.5K	R9	SYSTEM POWER TO ON & ATD TO ON
21	8	HI	3.2K to 3.4K	R33	ATD TO OFF & ACTION TO ON
19	5	HI	3.25K to 3.4K		
26	23	HI	< 10K		ATD TO ON
28	23	HI	< 10K		
23	26	HI	< 14K		
23	28	HI	< 14K		

Return the ATD and ACTION switches to OFF.

4.7.3.8 Power control. With the SYSTEM POWER switch ON, the SYSTEM POWER indicator must illuminate. Set the RADAR PWR switch to ON. The RADAR POWER indicator must illuminate. Return the RADAR PWR switch to OFF.

4.7.3.9 Fire voltage. Connect the ohmmeter, item 4, between TP45 and TP42. The ohmmeter reading must not be greater than 1.2 megohms.

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Set the FIRE RELAY switch to ON. Connect the ohmmeter, item 4, to TP42 and TP14. The ohmmeter must indicate 24.9 ±1 ohms. Return the FIRE RELAY switch to OFF and remove the ohmmeter leads. Connect the voltmeter, item 4, between TP42 (High) and TP45 (Low). Operate the GUN POWER switch to ON. The voltmeter must indicate +313.5 to +346.5 Vdc. Return the GUN POWER switch to OFF.

4.7.3.10 Functional circuits.

4.7.3.10.1 Sequence of operation. Perform the following steps.

a. ACTION switch to ON. The ACTION indicator must light.

b. Depress the TRIGGER switch momentarily. The GUN MOTOR HI, GUN MOTOR LO and FIRE VOLTAGE indicators must remain OFF.

c. RATE switch to HI and repeat step b.

d. GUN POWER switch to ON. The GUN POWER indicator must light.

e. ACTION switch to OFF.

f. GUN CLEAR switch to OPERATOR and hold.

g. ACTION switch to ON. The GUN MOTOR HI, ACTION, and GUN CLEAR indicators must light. The FIRE VOLTAGE indicator must remain OFF.

h. Release GUN CLEAR switch.

i. RATE switch to LO.

j. GUN CLEAR switch to OPERATOR and hold. The GUN MOTOR HI, ACTION, and GUN CLEAR indicators must light. The FIRE VOLTAGE indicator must remain OFF.

k. GUN CLEAR switch to AUTO.

l. Depress the TRIGGER switch and hold. The TRIGGER indicator, GUN MOTOR LO indicator and FIRE VOLTAGE indicator must light. The GUN MOTOR HI indicator must come on momentarily. The BURST LIMIT indicator must light after a short delay.

m. Release the TRIGGER switch. The GUN CLEAR indicator must come on momentarily.

n. Set the RATE switch to HI.

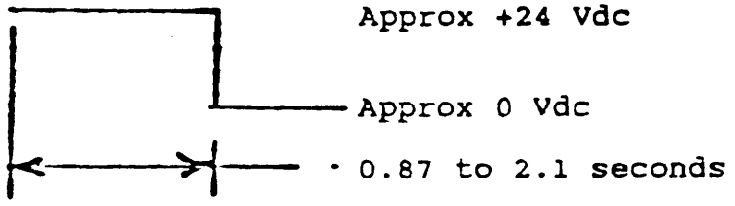
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o. Depress the TRIGGER switch and hold. The GUN MOTOR HI, TRIGGER and FIRE VOLTAGE indicators must light briefly then go out. When the GUN MOTOR HI indicator goes out, the GUN CLEAR and BURST LIMIT indicators must light. The GUN CLEAR indicator must go out after a delay.

p. Release the TRIGGER switch.

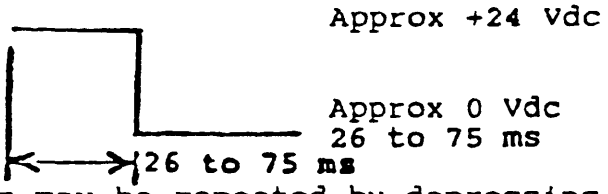
4.7.3.10.2 Time delays.

4.7.3.10.2.1 Burst limit. Connect the oscilloscope, item 5, y-axis to TP35. Set-up the oscilloscope sweep for single sweep, external positive triggering. Connect the external trigger to TP38. Depress and hold the TRIGGER switch. The oscilloscope must display the waveform shown below. Release the TRIGGER switch.



The waveform may be repeated by depressing and holding the TRIGGER switch. This sequence may be repeated as often as necessary.

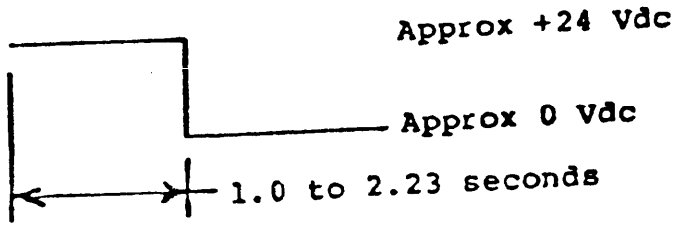
4.7.3.10.2.2 Gun motor high. Set the RATE switch to LO. Depress and hold the TRIGGER switch. The oscilloscope must display the waveform shown below. Release the TRIGGER switch.



The waveform may be repeated by depressing and holding the TRIGGER switch. This sequence may be repeated as often as necessary.

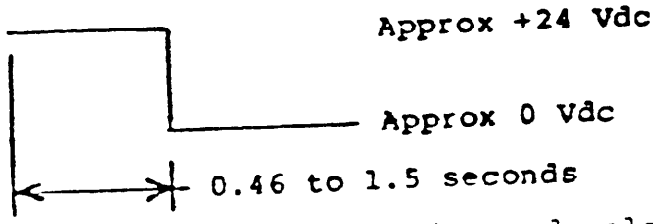
4.7.3.10.2.3 Clear. Connect the oscilloscope y-axis to TP37. Set-up the oscilloscope sweep for single sweep, external negative trigger. Connect the external trigger to TP47. Depress and release the TRIGGER switch. The oscilloscope must display the waveform shown below.

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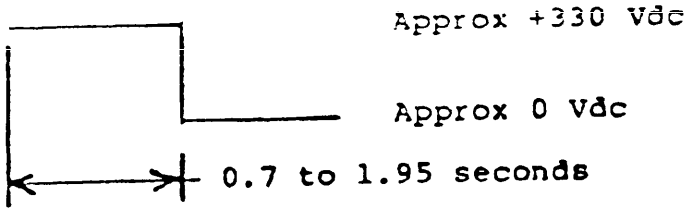
The waveform may be repeated by depressing and releasing the TRIGGER switch. This sequence may be repeated as often as necessary.

4.7.3.10.2.4 LO rate. Connect the oscilloscope y-axis to TP47. Connect the external trigger to TP38. Depress and release the TRIGGER switch. The oscilloscope must display the waveform shown below:



The waveform may be repeated by depressing and releasing the TRIGGER switch.

4.7.3.10.2.5 Fire voltage. Connect the oscilloscope y-axis to TP14 using a 100X probe, item 6. Connect the external trigger to TP47. Depress and release the TRIGGER switch. The oscilloscope must display the waveform shown below:



The waveform may be repeated by depressing and releasing the TRIGGER switch as often as necessary.

Remove the oscilloscope connections.

4.7.3.10.3 Hatch and elevation protection circuits. Perform the following steps:

a. Set the 5 VOLT switch to ON. Place RIGHT LIMIT switch to CLEAR. Observe that the LEFT LIMIT, RIGHT LIMIT and LOW LIMIT indicators are lit.

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b. Set the LEFT LIMIT and RIGHT LIMIT switches to the IN LIMIT position. The LEFT LIMIT, RIGHT LIMIT and LOW LIMIT indicators shall remain lit.

c. Set the 13 degrees and 18 degrees switches to the BELOW position. The LEFT LIMIT, RIGHT LIMIT and LOW LIMIT indicators shall remain lit.

d. Set the -5 degrees switch to the BELOW position. The LEFT LIMIT and RIGHT LIMIT indicators shall remain lit.

e. Set the HATCH switch to OPEN and LEFT LIMIT switch to CLEAR. Observe that only the LEFT LIMIT indicator is lit.

f. Set the LEFT LIMIT switch to IN LIMIT and RIGHT LIMIT switch to CLEAR. Observe that only the RIGHT LIMIT indicator is lit.

g. Set the -5 degrees switch to ABOVE and LEFT LIMIT switch to CLEAR. Observe that the LEFT LIMIT, RIGHT LIMIT and LOW LIMIT indicators are lit.

h. Set the 13 degrees switch to ABOVE and LEFT LIMIT switch to IN LIMIT. Observe that the LEFT LIMIT and RIGHT LIMIT indicators are lit.

i. Set the LEFT LIMIT switch to CLEAR and RIGHT LIMIT switch to IN LIMIT. Observe that the LEFT LIMIT and RIGHT LIMIT indicators remain lit.

j. Set the RIGHT LIMIT switch to the CLEAR position. Observe that the LEFT LIMIT, RIGHT LIMIT and LOW LIMIT indicators are lit.

k. Set the 13 degrees switch to BELOW and LEFT LIMIT switch to IN LIMIT positions. Observe that only the RIGHT LIMIT indicator is lit.

l. Set the LEFT LIMIT switch to CLEAR and the RIGHT LIMIT switch to IN LIMIT. Observe that only the LEFT LIMIT indicator is lit.

m. Set the LEFT LIMIT switch to the IN LIMIT position and observe that no indicators are lit.

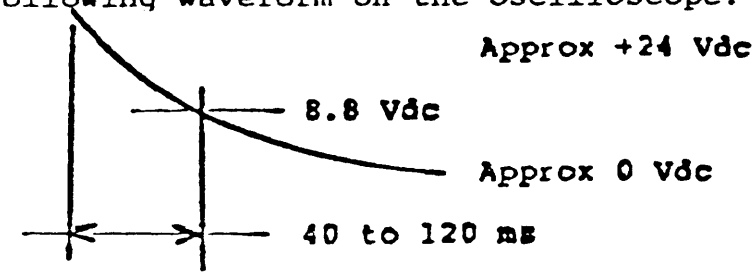
n. Set the 13 degrees switch to the ABOVE positions and observe that the LEFT LIMIT and RIGHT LIMIT indicators are lit.

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o. Set the 18 degrees switch to the ABOVE position and obser that the LEFT LIMIT, RIGHT LIMIT and LOW LIMIT indicators are lit.

4.7.3.11 Acquisition time delay relay. Set the ATD switch to ON. Observe that the ATD indicator lights. Return the ATD switch to OFF. Also, set the 5 VOLT switch to OFF.

4.7.3.12 Pulse generator. Connect the oscilloscope, item 5, y-axis to TP16. Set-up the oscilloscope sweep for single sweep, internal positive triggering. Operate the COUNTER PULSE switch and observe the following waveform on the oscilloscope.



The waveform may be repeated by releasing and depressing the COUNTER PULSE switch as often as necessary. Pause briefly in the CHARGE position to provide time for the capacitor in the D-box to fully charge.

Turn off the SYSTEM POWER switch. Using the volt-ohmmeter, item 4, measure the resistance between TP15 and TP17. The ohmmeter shall indicate 49.9 ±0.5 ohms.

4.7.3.13 Filter. Connect the 50 ohm output of the audio oscillator, item 7, to TP9 with the return to TP1. Also, connect the y-axis of the oscilloscope, item 5, to TP9. Adjust the audio oscillator for 1.0 KHz. Adjust the scope sweep to display the 1.0 KHz waveform. Set the audio oscillator output to provide 1.0 volt peak-to-peak as read on the oscilloscope. Move the oscilloscope y-axis connection to TP6 and observe that the oscilloscope indicates a peak-to-peak waveform of .09 to 0.2 volts.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in 3.7.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and markings shall be in accordance with Packaging Data Sheet PS8438325. The level of protection shall be as specified in the procurement document.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The D-box is a part of the M163A1 and M167A1 Weapon System. The D-box distributes 24 Vdc battery power throughout the mount, provides main power disconnect and overload protection. It also provides time delays and a 330 Vdc firing voltage. The D-box contains meters for monitoring auxiliary battery charging voltage and current.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. When first article is required (see 3.6).
d. Packing requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certification of conformance for each lot of shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment design for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Drawing. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by and identified as US Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Subject term (Key word) listing.

Vulcan Air Defense System (VADS)
Fire Control System

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 6110-A301)

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME US ARMY ARDEC STANDARDIZATION OFFICE	b. TELEPHONE (*Include Area Code*) (1) Commercial: (201) 724-6662 (2) AUTOVON: 880-6662
c. ADDRESS (*Include Zip Code*) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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